

12	C	Option A: The variable x is denoted as the distance <i>above the surface of the earth</i>. It is not defined from the centre of the Earth. Option B: $F_g = -\frac{dE_p}{dx}$. The force is the gradient of the $E_p - x$ graph, not ratio of E to x. [whereas the resistance is the ratio V to I, and not the gradient as given by dV/dI.] Option C: $F_g = -\frac{dE_p}{dx}$. Take note, for small distances above the Planet's surface, the gravitational field strength is approximately constant. Hence, the gradient of the graph is the same. Option D: The equation does not adhere to $F_g = -\frac{dE_p}{dx}$.
13	D	Frictional force by B on Q is the restoring force for Q. Without friction Q will not move as B slides